Controlled Production and Characterization of Chitosan Nanoparticles

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Traumatic Brain Injury and the Brain Patch

The goal of our lab is to develop a cell delivery system which mitigates the damage caused by secondary traumatic brain injury. The Brain Patch consists of a collagen scaffold seeded with mesenchymal stem cells, and chitosan nanoparticles (CNPs) (see Fig. 1).

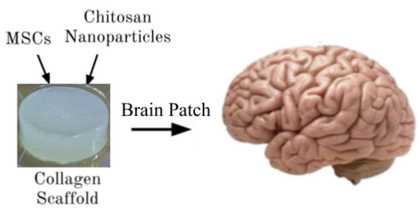


Figure 1. Brain Patch overview

Chitosan nanoparticles display several characteristics that are useful in a clinical context - they display high biocompatibility, excellent antimicrobial properties, and low immunogenicity[1].

The common method for generating chitosan-based nanoparticles is the ionic gelation of chitosan using a polyanionic sodium tripolyphosphate (TPP) crosslinker[2]. Another, less well investigated method of generating nanoparticles uses ultrasonic oscillators to generate nanoparticles[3].

In this study we aim to characterize the tunabilities of size and size distribution between the ultrasonic oscillation and ionic gelation methods.

Synthesis of Nanoparticles

Chitosan nanoparticles were generated in two ways, one by using ionic gelation and the other by using ultrasonic oscillators.

Gelation was performed by mixing two solutions together: one of 0.5 mg/ml of chitosan in 0.2 mg/ml acetic acid, and the other of varying amounts of TPP in water[4]. Size was tuned by varying TPP concentration.

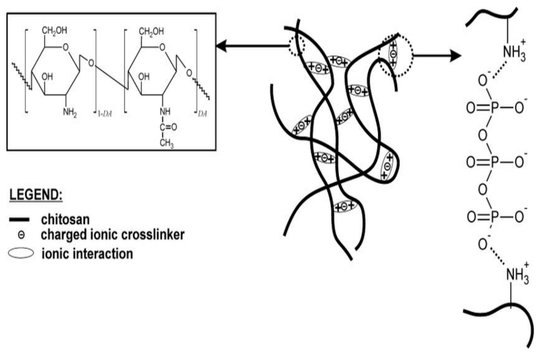


Figure 2. Cross linking of Chitosan using TPP.[5]

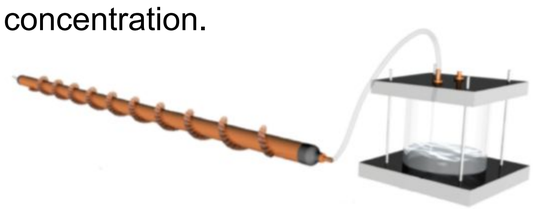


Figure 3. Atomization setup overview

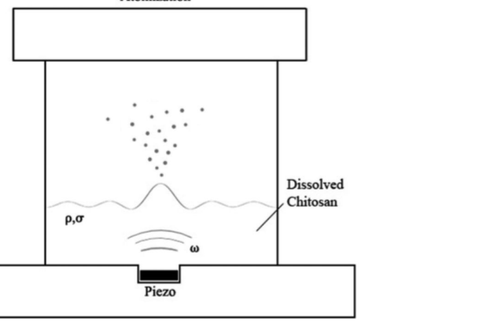


Figure 4. The aerosolization chamber

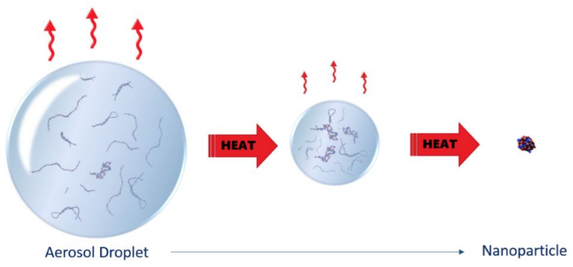


Figure 5. Generation of a single CNP

The ultrasonic oscillation method was carried out using piezo-electronic oscillators to aerosolize small droplets of chitosan polymer solution. The solvent was then evaporated off, leaving chitosan nanoparticles behind (Figs. 3, 4, and 5).

Sizing Crosslinked Nanoparticles

Dynamic Light Scattering(DLS) was used to size nanoparticles created by ionic gelation.

The effect of temperature on nanoparticle diameter was investigated. It was discovered that at temperatures of 4℃ and 70℃ ionically gelated nanoparticles have larger mean diameters (280nm and 340nm) and larger standard deviations (max = 100nm) whereas nanoparticles generated at room temperature had lower variability (std. Dev. = 30nm) and mean diameters (200nm) (Fig 6).

The effect of stirring speed was also investigated. CNP diameter showed an inverse relationship with stirring speed (Fig 7).

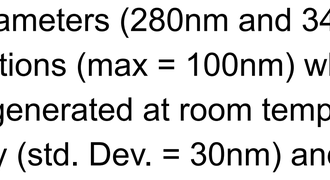


Figure 5. DLS Particle Size Measurement

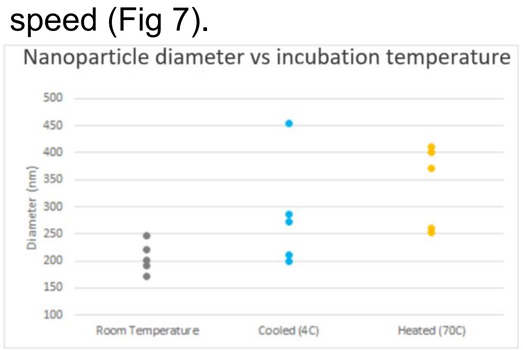


Figure 6. Nanoparticle diameter vs incubation temperature

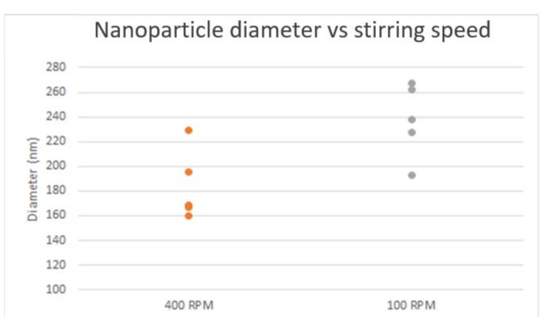


Figure 7. Ionically gelated Nanoparticle diameter vs stirring speed

Sizing Atomized Nanoparticles

Nanoparticles generated using ultrasonic atomization were sized using scanning electron microscopy (SEM) and ImageJ, an open source image processing software.

The effect of stock chitosan solution concentration on nanoparticle diameter was investigated. Nanoparticles had an average standard deviation of the mean diameter of 200nm, and a tunable average diameter between 300nm and 900nm (Fig. 8).

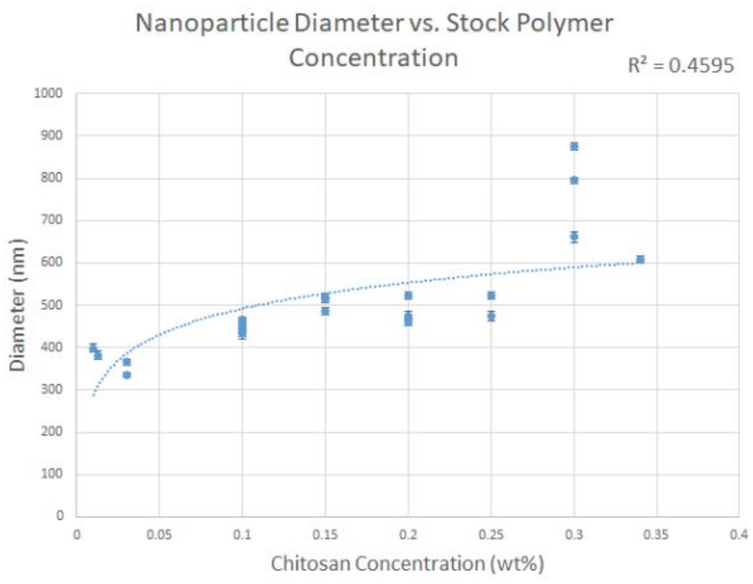


Figure 8. The effect of polymer solution concentration on nanoparticle diameter

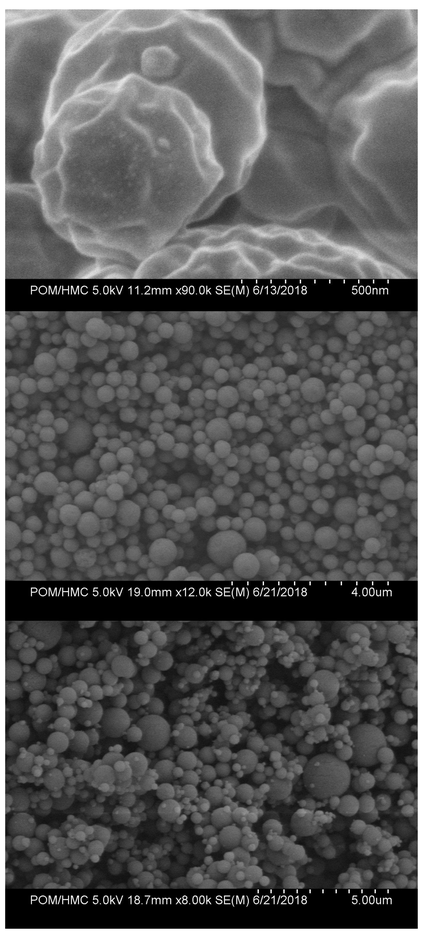


Figure 9. (Top) A single nanoparticle. (Center) Nanoparticles showing low size dispersity. (Bottom) Nanoparticles showing high size dispersity.

Discussion

The most striking result of this investigation was the difference in size profiles between the ultrasonic atomization and ionic gelation techniques. The oscillation method produced more polydisperse and larger nanoparticles than nanoparticles generated by ionic gelation.

Production Method	Ionic gelation	Ultrasonic Oscillation
Average Size range (nm)	150 - 450	300 - 900
Std. Dev. (nm)	30	200

Future Work

This research informs future work incorporating CNPs into a collagen gel to form a brain patch. If CNPs are to be loaded into a tissue engineering construct (Fig. 10), their physical dimensions must be known, as previous results have shown that nanoparticles of different sizes have different antimicrobial properties.

Future work includes characterizing the two production methods in terms of their efficacy regarding cell viability, antimicrobial and neuroprotective properties in order to produce the optimal nanoparticle for clinical use.

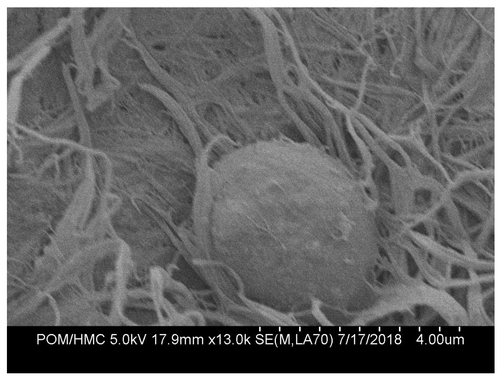


Figure 10: A nanoparticle in collagen gel

In addition to nanoparticle specific investigation, the effect of either of the two production methods on MSC differentiation needs to be understood, as well as their effects on collagen matrix stiffness, as matrix mechanical properties have been shown to affect stem-cell differentiation[6].

References

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